

Proposed Customized Automated Revit Toolset

Enhancing Productivity in Autodesk Revit

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What We'll Cover

01

Introduction & toolset Overview

Ribbon Detail

02

Script 1 — Audit Tools

Isolating exterior vs. interior walls with color overrides

03

Script 2 — Automatic Dimension Tool

Reading and writing element parameters programmatically

04

Script 3 — Automatic Export & Schedule Utilities

3D modelling and parametric design tools

05

Script 4 — Placement of Object By Co-ordinates

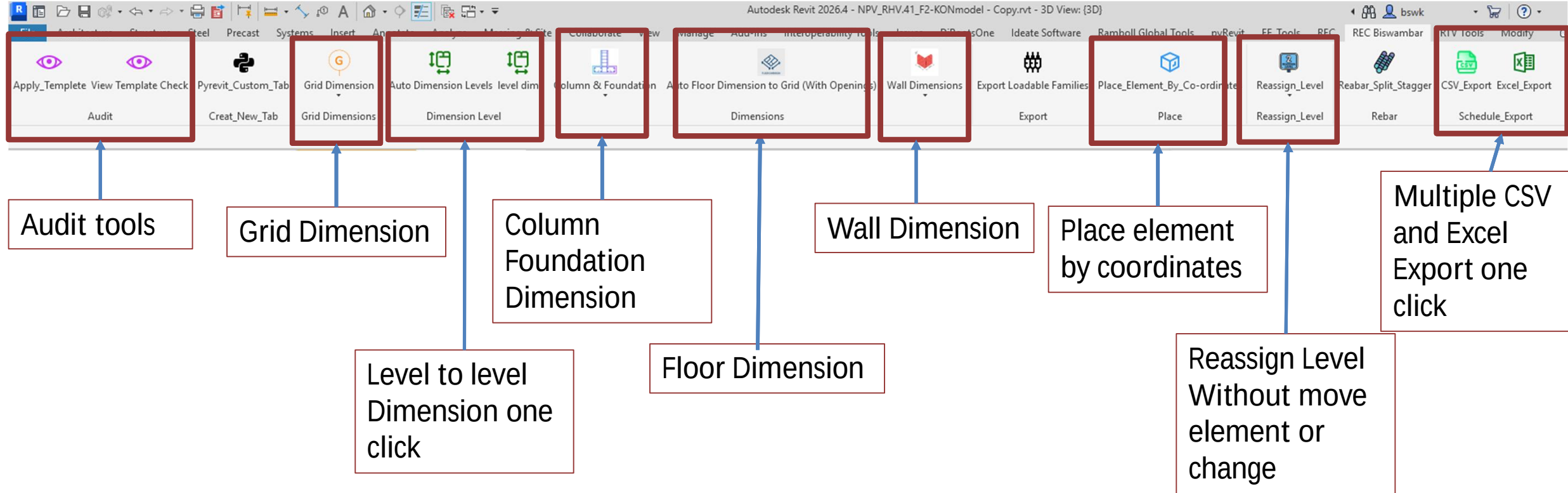
Geometry modelled in Rhino with layer-based organisation

06

Whats Next

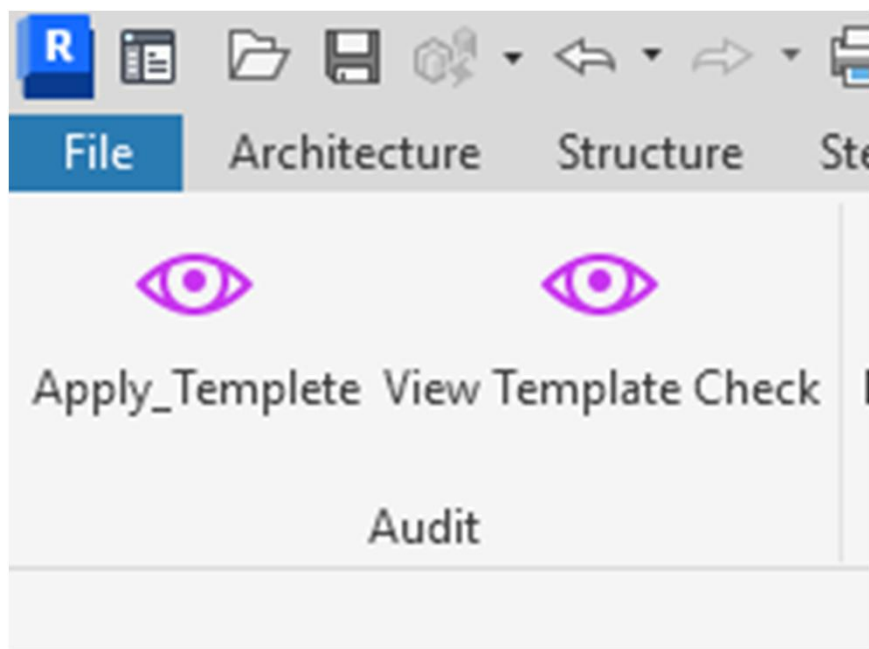
Converting Rhino geometry to 3D using parametric nodes

The REC Biswambar Ribbon



Audit Tools

- Apply Template – Apply standard view templates in bulk
- View Template Check – Validate template usage across views
- Ensures consistent graphics and discipline standards



View_Template_Check					
NPV_RHV.41A8_M96_H1_EXX_N9001	KON-Disponering - Bygning A+B - Plan03-05	B1 - Plan 04 Copy 1	✔ Applied	MKYA_Displaner	
NPV_RHV.41C_M96_H1_E00_N9001	KON-Disponering - Bygning C - Plan00	Disp - Planer	✘ NO TEMPLATE		
NPV_RHV.41C_M96_H1_E00_N9001	KON-Disponering - Bygning C - Plan00	B2 - Plan 00 Copy 1	✔ Applied	MKYA_Displaner	
NPV_RHV.41C_M96_H1_E01_N9001	KON-Disponering - Bygning C - Plan01	Disp - Planer	✘ NO TEMPLATE		
NPV_RHV.41C_M96_H1_E01_N9001	KON-Disponering - Bygning C - Plan01	B2 - Plan 01 Copy 1	✔ Applied	MKYA_Displaner	
NPV_RHV.41C_M96_H1_E02_N9001	KON-Disponering - Bygning C - Plan02	Disp - Planer	✘ NO TEMPLATE		
NPV_RHV.41C_M96_H1_E02_N9001	KON-Disponering - Bygning C - Plan02	B2 - Plan 02 Copy 1	✔ Applied	MKYA_Displaner	
NPV_RHV.41D_M96_H1_E00_N9001	KON-Disponering - Bygning D - Plan00	Disp - Planer	✘ NO TEMPLATE		
NPV_RHV.41D_M96_H1_E00_N9001	KON-Disponering - Bygning D - Plan00	B3 - Plan 00 Copy 1	✔ Applied	MKYA_Displaner	
NPV_RHV.41E_M96_H1_E00_N9001	KON-Disponering - Bygning E - Plan00	Disp - Planer	✘ NO TEMPLATE		
NPV_RHV.41E_M96_H1_E00_N9001	KON-Disponering - Bygning E - Plan00	B4 - Plan 00 Copy 1	✔ Applied	MKYA_Displaner	
NPV_RHV.80x_M65_H5_EXX_N5001	KON Fundamentsdetaljer	SEDN_Fundamentsskitser	✔ Applied	03. Sheet Views - Plan-Floor	
NPV_RHV.80x_M65_H5_EXX_N5001	KON Fundamentsdetaljer	SEDN_Fundamentsskitser Side 2	✔ Applied	03. Sheet Views - Plan-Floor	
NPV_RHV.80x_M65_H5_EXX_N5002	KON Principsnit	SEDN_Fundamentsskitser Side 4	✔ Applied	03. Sheet Views - Plan-Floor	
00.00.01	Section 1	Section 1	✘ NO TEMPLATE		
00.00.02	Section 2	Section 2	✘ NO TEMPLATE		

Audit Summary

Total Sheets Scanned: 47
Total Views Checked: 49

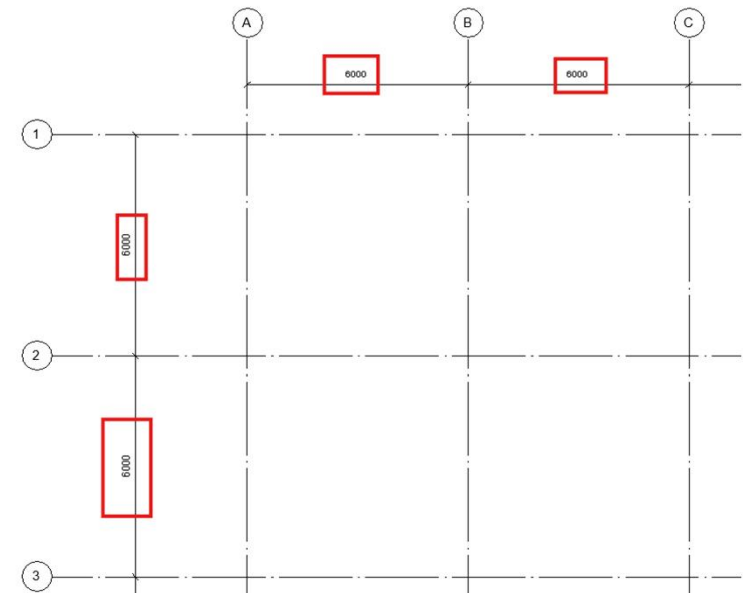
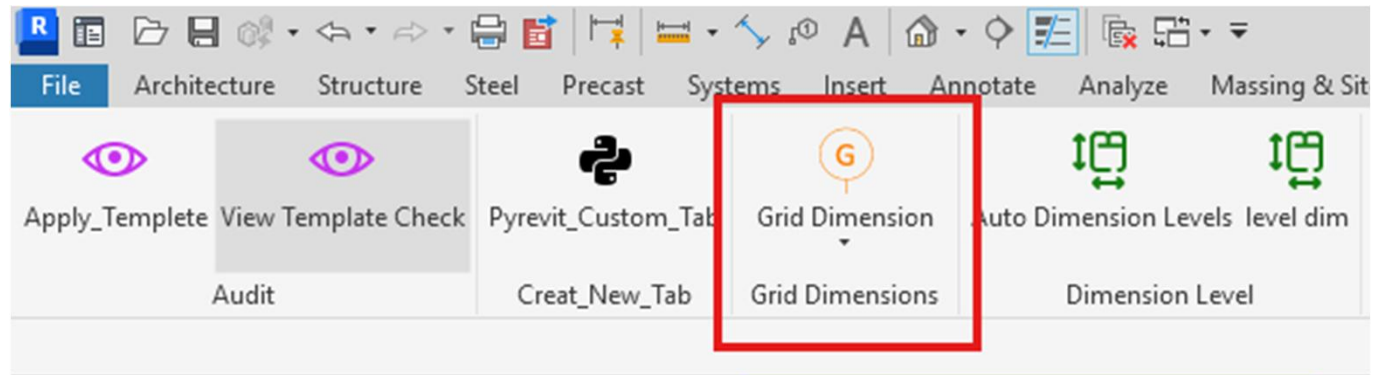
Template Usage Breakdown

Template: <None> | Count: 23
Template: MKYA_Floor Plan | Count: 9
Template: MKYA_Displaner | Count: 9
Template: MKYA_MV_Snit | Count: 5
Template: 03. Sheet Views - Plan-Floor | Count: 3

Overall Template Coverage: 53.1%

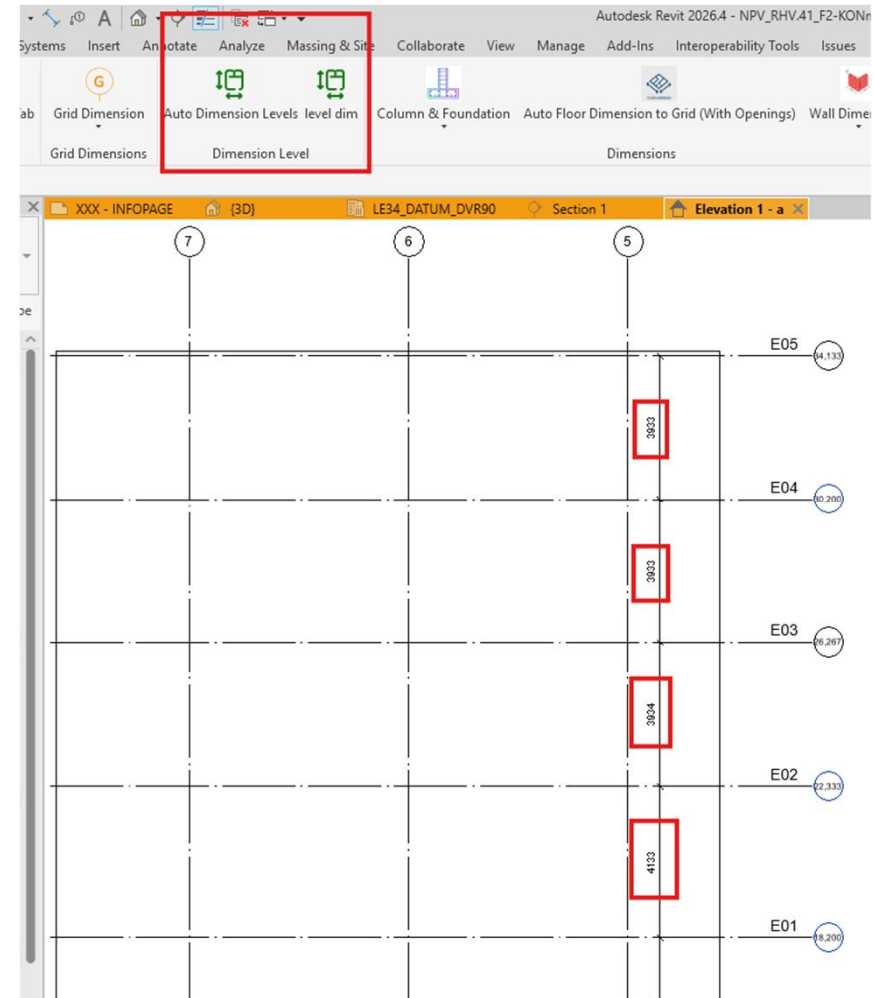
Grid Dimensioning

- Grid Dimension – Auto-create dimension strings along grids
- Eliminates manual click-by-click dimensioning
- Ensures consistent dimension placement



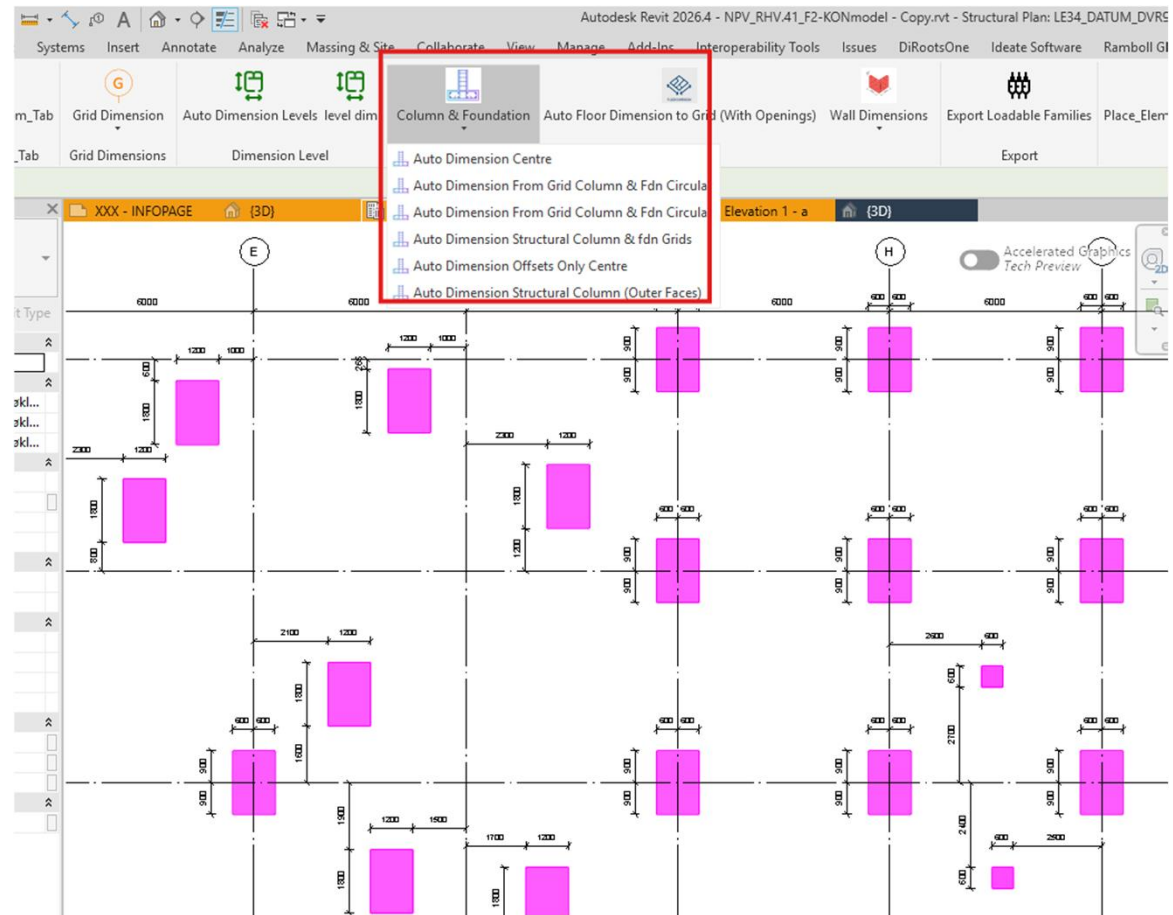
Level Dimensioning

- Auto Dimension Levels – Automatically dimension levels in sections/elevations
- Level Dim – Quick level-to-level dimension tool
- Saves hours in section drawing production



Smart Element Dimensioning

- Column & Foundation – Auto-dimension columns and footings to grids
- Auto Floor Dimension to Grid (with Openings) – Dimension slab edges and openings
- Wall Dimensions – Automated wall dimensioning



Smart Element Dimensioning- Workflow

Autodesk Revit 2026.4 - NPV_RHV.41_F2-KONmodel - Copy.rvt - Structural Plan: LE34_DATUM_DVR90

1 REC Biswambar

2 Wall Dimensions

3 Multiple

4 Select Plan Direction

5 Finish

6 View Range

7 Success: Plan view Wall Dimension generated perfectly with openings! Author-Biswambar Kshetri

Properties: Structural Plan: LE34_DATUM_DVR90

Graphics: View Scale: 1:100, Scale Value: 100, Display Model: Normal, Detail Level: Fine, Parts Visibility: Show Original, Visibility/Graphics Overrides: Edit..., Graphic Display Options: Edit..., Orientation: Project North, Wall Join Display: Clean all wall joins, Discipline: Coordination, Show Hidden Lines: By Discipline, Color Scheme Location: Background, Color Scheme: <none>, System Color Schemes: Edit..., Default Analysis Display: None, Sun Path: []

Underlay: Range: Base Level: None, Range: Top Level: Unbounded, Underlay Orientation: Look down

Extents: Crop View: [], Crop Region Visible: [], Annotation Crop: [], View Range: Edit..., Associated Level: LE34_DATUM_DVR90, Scope Box: None, Column Symbolic Offset: 304.8, Depth Clipping: Clip without line

Identity Data: View Template: <None>, View Name: LE34_DATUM_DVR90, Dependency: Independent, Title on Sheet, Referencing Sheet Collection: <None>, Referencing Sheet: 00.00.01, Referencing Detail: 1

Project Browser: Views (Ramboll Global 2 Levels)

- 01. User Modelling Views
 - MKYA-Disp-plan
 - MKYA-Snit
 - Section: B1 - Snit 1
 - Section: B1 - Snit A
 - Section: B1 - Snit B
 - Section: B2 - Snit 1
 - Section: B2 - Snit A
- SEDN-Disp-plan

- 03. Sheet Views
- Plan-Floor
 - Plan-Wall_Column_Beam(Down)
- Structural Plan: LE34_DATUM_D
- ???
- 3D View: [3D]
- Elevation: Elevation 1 - a
- Section: Section 1
- Section: Section 2
- Section: Section 3
- Section: Section 4

Legends

- Schedules/Quantities (Ramboll Global 1 lev)
- 02. Document List
- 03. Design
- ???
- Sheets (Ramboll Global 2 Levels)
- Families
- Groups
- Revit Links

Smart Element Dimensioning- Video

Autodesk Revit 2026.4 - NPV_RHV.41_F2-KONmodel - Copy.rvt - Structural Plan: LE34_DATUM_DVR90

File Architecture Structure Steel Precast Systems Insert Annotate Analyze Massing & Site Collaborate View Manage Add-Ins Interoperability Tools Issues DiRootsOne Ideate Software Ramboll Global Tools pyRevit EF_Tools REC Biswambar RTV Tools

Apply_Template View Template Check Pyrevit_Custom_Tab Grid Dimension Auto Dimension Levels level dim Column & Foundation Auto Floor Dimension to Grid (With Openings) Wall Dimensions Export Loadable Families Place_Element_By_Co-ordinates Reassign_Level Rebar_Split_Stagger CSV_Export Excel_Export

Audit Creat_New_Tab Grid Dimensions Dimension Level Dimensions Export Place Reassign_Level Rebar Schedule_Export

Properties

Structural Plan
SV-ST_PLAN-WALL_COLUMN_BEAM-DOWN_RDK

Structural Plan: LE34_DATUM_DVR90 Edit Type

Graphics

View Scale 1: 100
Scale Value 1: 100
Display Model Normal
Detail Level Fine
Parts Visibility Show Original
Visibility/Graphics Overri... Edit...
Graphic Display Options Edit...
Orientation Project North
Wall Join Display Clean all wall joins
Discipline Coordination
Show Hidden Lines By Discipline
Color Scheme Location Background
Color Scheme <none>
System Color Schemes Edit...
Default Analysis Display ... None
Sun Path ☐

Underlay

Range: Base Level None
Range: Top Level Unbounded
Underlay Orientation Look down

Extents

Crop View ☐
Crop Region Visible ☐
Annotation Crop ☐
View Range Edit...
Associated Level LE34_DATUM_DVR90
Scope Box None
Column Symbolic Offset 304,8
Depth Clipping Clip without line

Identity Data

View Template <None>
View Name LE34_DATUM_DVR90
Dependency Independent
Title on Sheet
Referencing Sheet Collec... <None>
Referencing Sheet 00.00.01
Referencing Detail 1

Project Browser - NPV_RHV.41_F2-KONmodel...

Views (Ramboll Global 2 Levels)

01. User Modelling Views

+ MKYA-Disp-plan

- MKYA-Snit

Section: B1 - Snit 1
Section: B1 - Snit A
Section: B1 - Snit B
Section: B2 - Snit 1
Section: B2 - Snit A

+ SEDN-Disp-plan

03. Sheet Views

+ Plan-Floor

Plan-Wall_Column_Beam(Down)

Structural Plan: LE34_DATUM_I

3D View: (3D)
Elevation: Elevation 1 - a
Section: Section 1
Section: Section 2
Section: Section 3
Section: Section 4

Legends

Schedules/Quantities (Ramboll Global 1 le

02. Document List

Sheet List KON

03. Design

Peletabel

Floor Material Takeoff
Structural Column Material Takeoff
Structural Framing Material Takeoff
Wall Material Takeoff

Sheets (Ramboll Global 2 Levels)

Families

Groups

Revit Links

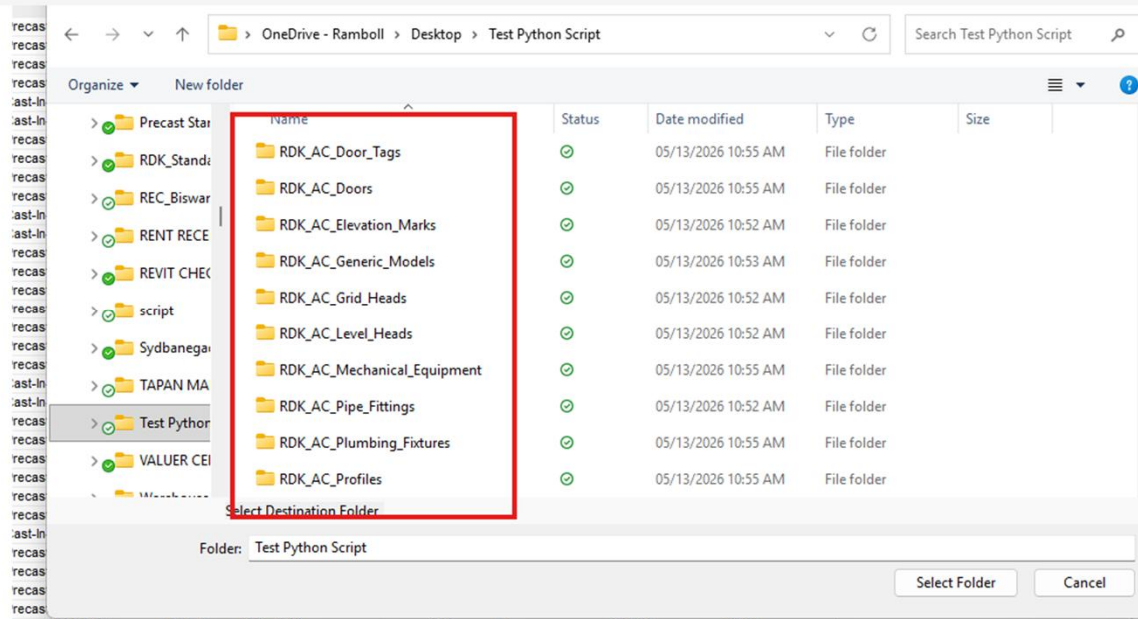
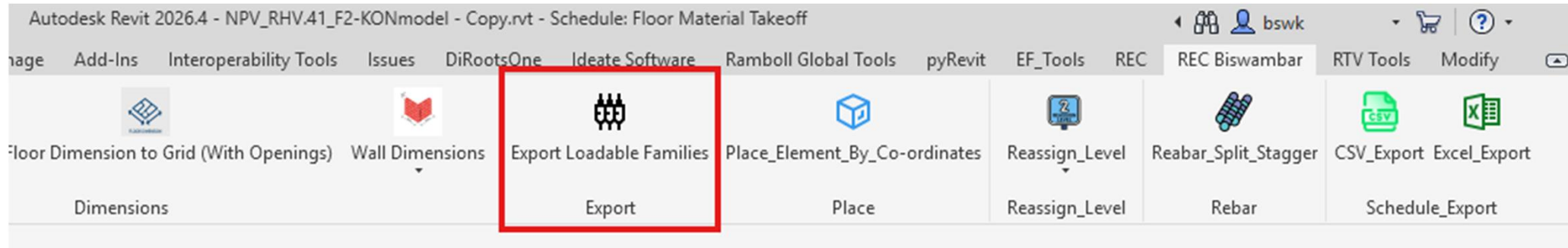
1: 100

Walls: Basic Wall: WAL_Precast-Interior-150_RDK

Main Model

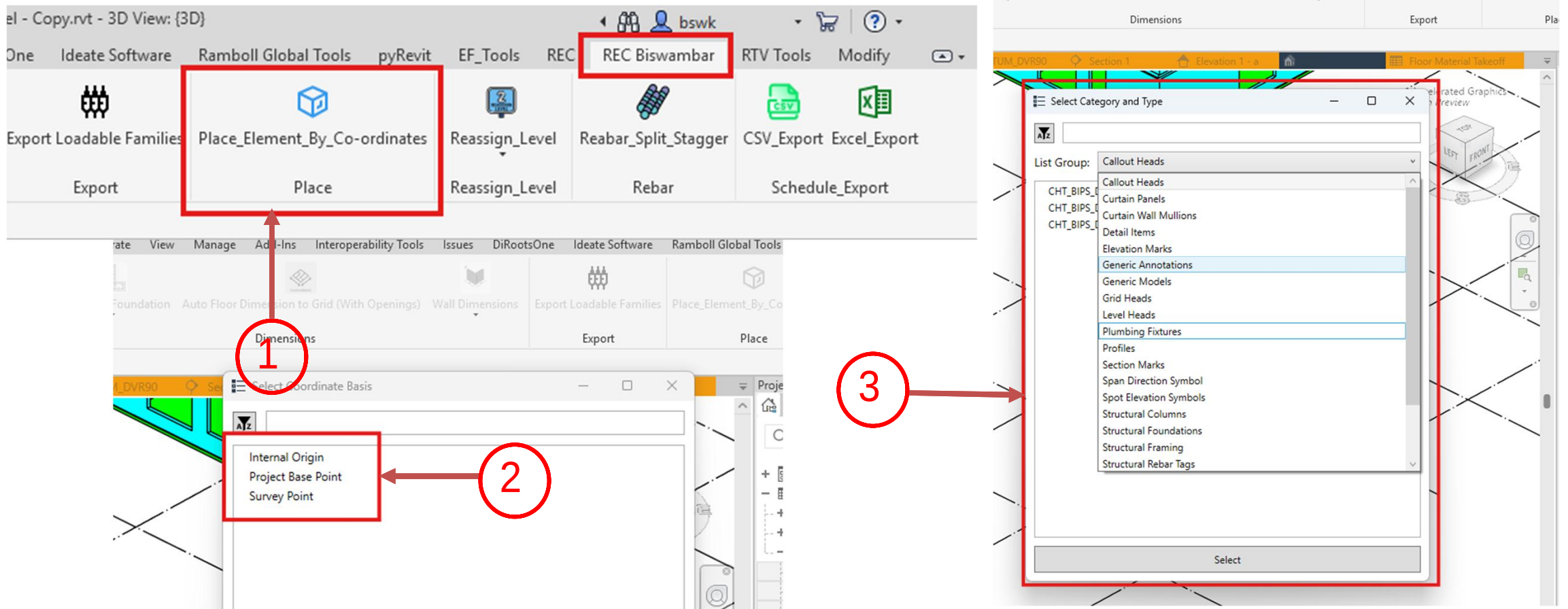
Export Panel

- Export Loadable Families – Batch-export all loadable families from the project
- Useful for family library management and project handover



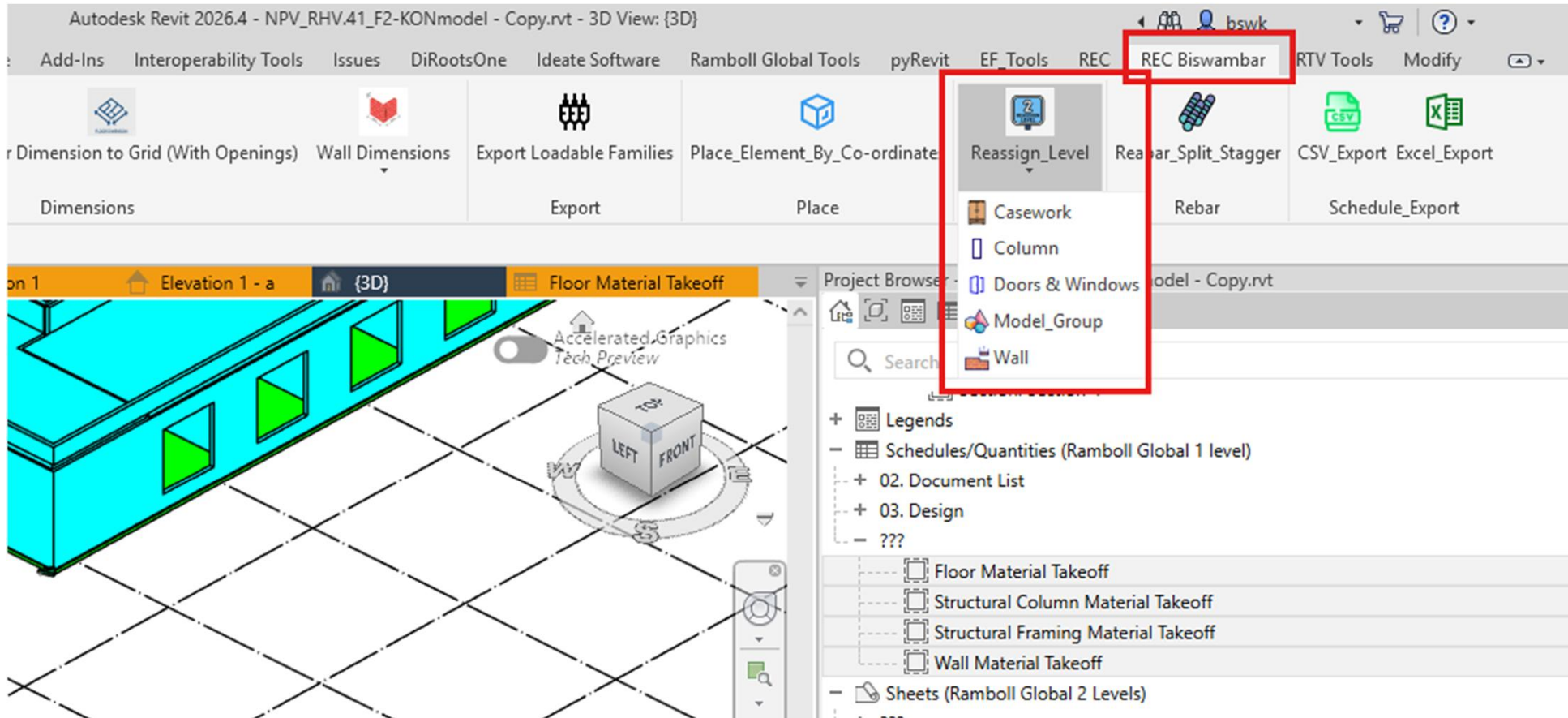
Coordinate-Based Placement

- Place Element by Co-ordinates – Place elements using X, Y, Z
- Useful for survey points, MEP fixtures, or pile layouts
- Reduces manual placement errors



Level Management

- Reassign Level – Move elements between levels while preserving their position
- Fixes common modeling errors quickly
- Prevents geometry shift during level changes



Data Export

- CSV Export – Export schedules to CSV format
- Excel Export – Direct export to Excel with formatting preserved
- Enables data analysis, BOQ preparation, and reporting

Autodesk Revit 2026.4 - NPV_RHV.41_F2-KONmodel - Copy.rvt - Schedule: Floor Material Takeoff

Systems Insert Annotate Analyze Massing & Site Collaborate View Manage Add-Ins Interoperability Tools Issues DiRootsOne Ideate Software Ramboll Global Tools pyRevit EF_Tools REC REC Biswambar RTV Tools Modify

Grid Dimension Auto Dimension Levels level dim Column & Foundation Auto Floor Dimension to Grid (With Openings) Wall Dimensions Export Loadable Families Place_Element_By_Co-ordinates Reassign_Level Rebar Split Stagger CSV Export Excel Export Schedule Export

XXX - INFOPAGE {3D} LE34_DATUM_DVR90 Section 1 Elevation 1 - a {3D} Floor Material Takeoff

<Floor Material Takeoff>

A	B	C	D	E	F
Family and Type	Type	Default Thickness (	Count	Area (m²)	Material Volume (m³)
Floor: FLO_Precast-270_RDK	FLO_Precast-270_RDK	270	1	46,14	12,46
Floor: FLO_Precast-320_RDK	FLO_Precast-320_RDK	320	1	36,10	11,55
Floor: FLO_Precast-320_RDK	FLO_Precast-320_RDK	320	1	629,24	201,36
Floor: FLO_Precast-270_RDK	FLO_Precast-270_RDK	270	1	253,30	68,39
Floor: FLO_Cast-In-Place-120_RDK	FLO_Cast-In-Place-120_RDK	120	1	313,49	37,62
Floor: FLO_Cast-In-Place-120_RDK	FLO_Cast-In-Place-120_RDK	120	1	209,66	25,16
Floor: FLO_Precast-320_RDK	FLO_Precast-320_RDK	320	1	81,40	26,05
Floor: FLO_Precast-320_RDK	FLO_Precast-320_RDK	320	1	573,78	183,61
Floor: FLO_Precast-320_RDK	FLO_Precast-320_RDK	320	1	81,40	26,05
Floor: FLO_Precast-320_RDK	FLO_Precast-320_RDK	320	1	573,78	183,61
Floor: FLO_Cast-In-Place-120_RDK	FLO_Cast-In-Place-120_RDK	120	1	126,99	15,24
Floor: FLO_Cast-In-Place-120_RDK	FLO_Cast-In-Place-120_RDK	120	1	97,56	11,71
Floor: FLO_Precast-320_RDK	FLO_Precast-320_RDK	320	1	81,40	26,05
Floor: FLO_Precast-320_RDK	FLO_Precast-320_RDK	320	1	573,78	183,61
Floor: FLO_Precast-180_RDK	FLO_Precast-180_RDK	180	1	93,23	16,78
Floor: FLO_Precast-180_RDK	FLO_Precast-180_RDK	180	1	608,04	109,45
Floor: FLO_Precast-270_RDK	FLO_Precast-270_RDK	270	1	143,81	38,83
Floor: FLO_Precast-270_RDK	FLO_Precast-270_RDK	270	1	107,44	29,01
Floor: FLO_Precast-270_RDK	FLO_Precast-270_RDK	270	1	320,79	86,61
Floor: FLO_Cast-In-Place-120_RDK	FLO_Cast-In-Place-120_RDK	120	1	545,89	65,51

Project Browser - NPV_RHV.41_F2-KONmodel - Copy.rvt

Search

Legends

Schedules/Quantities (Ramboll Global 1 level)

02. Document List

03. Design

Floor Material Takeoff

Structural Column Material Takeoff

Structural Framing Material Takeoff

Wall Material Takeoff

Sheets (Ramboll Global 2 Levels)

???

H0 General

H1 Plan

H2 Elevation

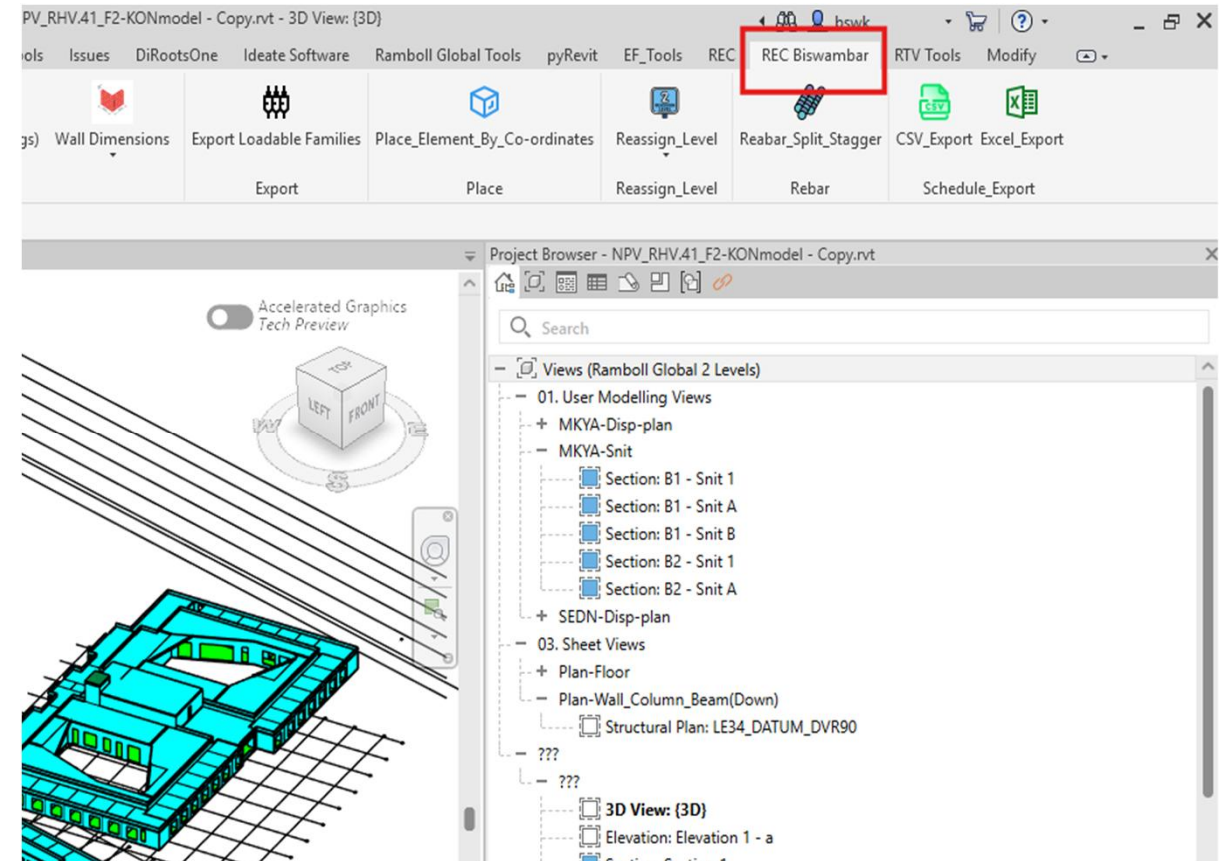
H3 Section

H4 Building Section

H5 Detail

Why Use REC Customized User Tab?

- Time savings: Up to 60–70% on dimensioning tasks
- Consistency: Standardized output across projects
- Error reduction: Automated placement & checks
- Scalability: Built on pyRevit — easy to extend
- Team enablement: Empowers users to focus on design, not drafting



Where It Adds Value

- Structural projects with complex grids
- Foundation & pile layout drawings
- Project handover & family library creation
- QA / model audits before milestones

What's Next?

- Integration with Ramboll Global Tools
- Additional dimensioning automation for MEP
- Cloud-based settings sync across teams
- User feedback channel for new feature requests

Thank You

Questions & Discussion

[Your Name]
[Your Email] | [Your Phone]
Ramboll – [Your Office]